

A conducive ecosystem is paramount for faster commercialisation of the drone technology

The draft policy 2.0 rules for manufacturing and use of drones in India necessitate for the authorised drone manufacturer or importer to sell their drone equipment only to an individual or entity approved by the DGCA. The industry stakeholders are further expecting the government to set up more and more test facilities for start-ups, facilitating knowledge sharing among the entrepreneurs, academia, and the large enterprises with world-class R&D infrastructure capability. Government bodies can ensure a smooth and accelerated commercialisation of drones through faster approval of intellectual property (IP) rights.

The Drone Policy 2.0 seeks to provide a way for foreign players to invest in India. It is expected to create a conducive ecosystem that can attract foreign investment in Indian drone market. While the draft talks about allowing 100% foreign direct investment (FDI) in UAS-based commercial civil aviation services, the current importing and permission framework are required to be revamped in order to make the 'Make in India' an attractive alternative.

"The focus is on creating 'Make in India' friendly environment to attract foreign manufacturers. We intend to push relentlessly for 'Make in India, Make for the World' under the Atmanirbhar Bharat Abhiyan."

Amber Dubey,

Joint Secretary, Ministry of Civil Aviation

more entrepreneurs join the bandwagon.

Though the government has not made any testing certification mandatory for commercial drones, Quality Council of India (QCI) and renowned Qualification Bodies (QB) are working with the government to develop a mechanism to provide quality certification to the drones. Additionally, the Ministry of Defense (MoD) is setting up a cluster of testing infrastructure for military equipment, including drones. With an allocated budget of four billion rupees, the proposed testing centres will be run by a private consortium

from industry, planned to be set up in two defense corridors in UP and Tamil Nadu. Such testing centres can very well facilitate standardising the specifications and performances of commercial drones by enabling the manufacturers, service providers, software developers, sensor and other sub-system level manufacturers to test and certify their products before they enter the market.

Group Captain Jagtreshwar Singh, NSCS suggests to bring efficiency in the process of R&D by creating calibrated test facilities. He adds, "We should not waste time and resources in developing technology

which is already developed. It will make sense to involve the drone users at the R&D stage itself. Also, it is crucial to test the parameters for flight testing on-field. Gather the information from the users and leverage the real-life experiences of the individuals." He further adds, "there has to be harmony among the labs. For example, IIT Kanpur and other government labs should facilitate knowledge sharing, thus making the development work collaborative across the ecosystem to avoid duplication of work and bring more efficiency."

Prof. Agrawal and Prof. Ghosh, IIT Kanpur:

"We still have many miles to go. Apart from IIT Kanpur, more such academic institutions need to participate in this sector. We request to government to create a proper eco-system. The labs should be able to issue certificates, so the industry can get their innovations certified, which is valid for the next level of certifications, for example, DGCA."

Defense Research and Development Organization (DRDO)

Unmanned Aerial Vehicle (UAV) or Unmanned Aircraft Systems (UAS) programme:

- **DRDO Lakshya.** It was developed for discreet aerial reconnaissance and target acquisition.
- **DRDO Nishant.** It has completed its developmental phase and user trials. Though primarily designed for intelligence-gathering application by the defense, now it is finding its application in reconnaissance, training, surveillance, target designation, artillery fire correction, and damage assessment.
- **DRDO Rustom.** It has been designed in line with the American Predator UAV. The Rustom is a Medium-Altitude Long-Endurance (MALE) system, which acts like the Predator, to be used for both reconnaissance and combat missions.

Manufacturing of drone equipment (OEMs/ODMs) expected to gain momentum post the standardisation

The Ministry of Civil Aviation has clearly set the guidelines for manufacturing drones which comply with the NPNT (No Permission-No Take off) and Data Protection & Privacy Policy. Additionally, the UAV equipment also need to meet the requirement of airworthiness, to be built around specific hardware standards and bimodal control features, allowing air traffic to take control of a drone in BVLOS flight. The Indian Drone Policy should be able to drive global standardisation of hardware and testing by working on airworthiness standards encompassing specific hardware